An V. Nguyen

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Research orientations

Large foundation models: Sequence processing, Parameter-efficient fine-tuning, Efficient models

Education

Korea University *Seoul, South Korea*

M.S. in Artificial Intelligence (*Taught in English*) *Sep 2024 - Feb 2027*

- GPA: 4.5/4.5 – based on standard 24 credits
- Graduation thesis: New paradigms in Parameter-efficient Fine-tuning for Large foundation models

Hanoi University of Science and Technology — HUST *Hanoi, Viet Nam*

Engineering degree: Control Engineering and Automation (5-year program) *Sep 2015 - Jan 2020*

- GPA: 3.7/4.0. Ranked: 5thAI / 299 in the major cohort
- On average, fewer than 2% of HUST students graduated with a GPA of 3.6/4.0 or higher
- Research project: Vector control specialised for switched reluctance motors
  Grade: A+ (9.9/10)

Research Experience

Stochastic Dynamics and Machine Learning Lab, Korea University *Seoul, South Korea*

Master's student – Research assistant. Advisor: *Prof. Anh Tong* *Aug 2024 - present*

- Contributed to a funded National Research Foundation of Korea proposal, 2025.

Center for Technology Innovation – CTI Lab, HUST *Hanoi, Vietnam*

Undergraduate student – Research intern. Advisor: *Prof. Minh C. Ta* *Mar 2018 - Feb 2020*

Publications and Patents

C2 **An Nguyen** and Anh Tong, "SaMA: Morpho Adaptation via Asymmetric Expansion of Kronecker Product", *Conference on Neural Information Processing Systems (NeurIPS)*, 2026.

C1 **An Nguyen**, Jaesik Choi and Anh Tong, "LoCO: Low-rank Compositional Rotation Fine-tuning", *International Joint Conference on Artificial Intelligence (IJCAI)* (acceptance rate 13.6%), 2026.

W1 **An Nguyen** and Anh Tong, "Sparse Structured Matrices: Efficient Adapter Rank in Fine-tuning Foundation Models", *ICML* workshop*: Continual Adaptation at Scale*, **Oral Presentation**, 2026.

P1 [*Patent*] **Nguyen Van An** and Anh Tong, "Method and Apparatus for Weight Parameterization of Neural Network Using Sparse Structured Matrix Decomposition", *S. Korea Patent Application* No. 10-2026-0160393, filed 2026.

Awards / Honors / Grants

ICML Registration Grant (Awarded by the Organizing Committee of ICML workshop CATS) 2026

Travel Grant (IJCAI-ECAI 2026) 2026

Foreign Global Leader Scholarship, Korea University Spring 2025, 2026

Foreign Natural Sciences and Engineering Scholarship, Korea University Fall 2024, 2025

Software Competency Certificate, LG Electronics Development Vietnam 2024

- *LG corporation's coding and algorithms assessment; one of only 3 LGEDV developers to pass that round.*

The Excellence Scholarship, HUST 2015 - 2017

Selected Projects

Linear Minimization Oracle (LMO) and optimization in large model adaptation *2026 - Present*

- Investigating how LMO-based optimizers interact with matrix factorization, designing a new fine-tuning adapter

Structured sparse matrix: Morpho parameterization in fine-tuning large models *2025 - Present*

- Proposed SaMA: sparse adapters generalizing the Kronecker product, with a quadratic rank property
- Showed its gauge symmetry collapses from LoRA's $GL(r)$ to diagonal scalings, eliminating coupling effect
- Matches or approaches full fine-tuning performance on complex tasks such as math and reasoning

Low-rank compositional rotations in orthogonal fine-tuning of large models *2024 - 2026*

- Proposed low-rank skew-symmetric rotations composed in parallel, without materializing $d \times d$ matrices
- Achieved superior or competitive on LLMs, ViTs, and diffusion models compared to other orthogonal methods

Advanced control techniques on electric motor drives (EE) *2018 - 2020*

- Investigated vector control in electric vehicles and reluctance motors. Proposed new control strategies

[Personal] Reading application: Quick dictionary lookup as *an Edge extension* *2026*

Skills

Programming Python/PyTorch, C/C++, C#, MATLAB/Simulink

Languages English: IELTS 7.5 overall (Sep 2026); TOEIC 920. Vietnamese: Native

Academic Service

Student Volunteer: IJCAI-ECAI (Germany) Aug 2026

Sub-reviewer: ICLR, IJCAI, AAAI, NeurIPS 2025, 2026

Professional Experience

LG Electronics Development Vietnam – *Automotive software developer* *Jun 2023 – Aug 2024*

Vietnam-Korea Institute of Science and Technology – *R&D engineer in motor drive* *Mar 2021 – Jun 2023*

Viettel Aerospace Institute – *R&D engineer in control systems* *Mar 2020 – Mar 2021*

References

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